

Q1.

Vibrational frequency for the C=O stretching:

CH₃CONH₂: 2139.19 Hz

CH₃COCH₃: 2123.22 Hz

CH₃CHO: 2113.01 Hz

CH₃COCl: 2141.52 Hz

The trend of frequencies is:

CH₃COCl > CH₃CONH₂ > CH₃COCH₃ > CH₃CHO

The reason why the trend is like this is because of multiple factors, including steric hindrance on the C=O bond; and the mass of the R-C-R/R-C-H group on the other end of the vibration (less important).

Q2.

The steps that are to be taken when running an MD simulation include:

1. Downloading the PDB file. Removing unnecessary crystals, and recreating a pdb file. Then using the TOP and the PAR file, make the PSF file (here we use the package psfgen to make the PSF file in Vmd).
2. The PSF file is to be loaded with the data of the PDB file, and then solvated with the water box to form the final pdb and psf file with the water box.
3. The config file is configured according to the requirements
4. A simulation is run on namd2 from the config file, which also automatically contains all relevant information, regarding timestep, total steps, the path of the psf and pdb file too.
5. The data is analysed accordingly.

